MODULE 1 ASSIGNMENT

Home Wi-Fi Router, Network Switch, Ports, LEDs and Gaming Café Network

Student Name	Dhwanit
Subject	Information Technology / Computer Networking
Assignment Topic	Router and Switch Hardware

Assignment objective: Learn to identify common ports and status indicators on routers and switches, understand how the two devices handle network traffic, and plan a simple wired network for a small gaming café.

Includes original virtual hardware diagrams, comparison tables, labelled port explanations, and a practical network scenario.

Introduction

Routers and switches are common networking devices, but they do different jobs. A home Wi-Fi router normally connects a local home network to the internet and often includes Wi-Fi, a small Ethernet switch, DHCP, and basic firewall features. A network switch connects wired devices to the same local area network (LAN), such as desktop computers, printers, servers, and wireless access points.

The exact shape and port arrangement depend on the brand and model. A WAN port is usually used for the incoming internet connection on a home router. LAN ports connect local devices. Some business routers include a console port for local management; many ordinary home routers do not have one. Status LEDs indicate whether the device is powered on and whether network links or Wi-Fi are active.

1. Labelled Diagram of a Typical Home Wi-Fi Router

VIRTUAL DIAGRAM — TYPICAL HOME WI-FI ROUTER (REAR / TOP VIEW)

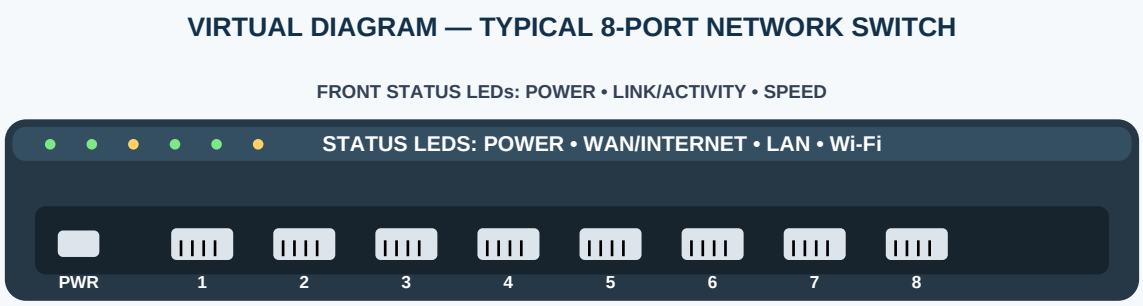
Illustrative layout only: exact port positions and colours vary by manufacturer.

WAN = incoming internet link | LAN = local devices | Power = adapter

Router ports and indicators

Label	Location / appearance	Purpose
Power port	Usually on the rear panel, near the power button or reset opening.	Connects the approved power adapter to supply electricity to the router.
WAN / Internet	Usually a separate Ethernet port on the rear; may be coloured differently.	Connects the router to the modem or ONT so traffic can reach the ISP.
LAN 1–4	A group of Ethernet ports on the rear panel.	Connects local wired devices such as a PC, TV, printer, or switch.
Status LEDs	Often on the front or top edge; may be small lights or icons.	Show power, internet/WAN, Ethernet link/activity, and Wi-Fi status.
Reset	Often a small recessed button or pinhole on the rear.	Restores factory settings when used according to the device instructions. It is not a normal on/off switch.

2. Labelled Diagram of a Typical Network Switch



Illustrative layout only: exact port positions and colours vary by manufacturer.

Numbered ports connect wired devices; uplink may use any port on a basic switch.

Switch ports and indicators

Label	Location / appearance	Purpose
Power port	Usually at the rear or side, depending on the model.	Connects the switch to its power adapter or power supply.
Ethernet ports 1–8	Usually arranged in a row on the front or rear panel.	Connects computers, printers, servers, access points, or an uplink to another network device.
Link/activity LEDs	Often located above or below each Ethernet port, or in the port itself.	A light commonly means a link is established; blinking often indicates data activity. Exact colours vary by model.
Power LED	Usually near the power port or on the front panel.	Indicates that the switch is receiving power and has started.
Console port	Present on many managed business switches, often a serial or USB console port.	Allows an administrator to configure or troubleshoot the switch locally. Basic unmanaged home switches usually do not have one.

Note about the diagrams: These are virtual educational diagrams, not photographs of a particular product. Real routers and switches differ by manufacturer. Always check the labels printed beside the ports and the manual for the exact model before connecting cables.

3. Router vs. Switch: Three Key Differences

The most important difference is that a router connects separate IP networks, while a typical Layer 2 switch connects devices inside the same LAN. Both forward traffic, but they use different information to make forwarding decisions.

Difference	Router	Switch
1. Main function	Connects different networks—for example, a home LAN to the internet.	Connects multiple devices within one local network, such as eight PCs in a lab.
2. Addresses used	Uses IP addresses and routing tables to decide where packets should go. A home router may also use NAT to share one public internet connection.	A typical Layer 2 switch learns MAC addresses and forwards Ethernet frames toward the port associated with the destination MAC address.
3. Traffic and network boundaries	Usually acts as the gateway between the private LAN and the ISP network; it can apply routing rules and firewall policies.	Normally forwards local Ethernet traffic within a LAN. A basic switch does not usually route between IP networks or provide internet access by itself.

A quick example

If a laptop wants to reach a website, the router forwards the traffic from the home network toward the ISP. If one gaming PC sends data to another device on the same LAN, a switch can forward the Ethernet frames locally. The router and switch may both be involved when the destination is outside the local network.

Extra point: Some advanced Layer 3 switches can route between IP networks, and many home routers contain a built-in switch. The comparison above describes a typical home router and a typical Layer 2 access switch.

4. Identifying Ports and LEDs on Real or Virtual Images

For this assignment, the two labelled virtual diagrams on the previous pages identify the power port, Ethernet ports, and status indicators. Use the following descriptions to explain their position and purpose. If your teacher specifically requires photographs, take photos of available equipment or use a properly credited image from a suitable source.

Router image identification

• Power port: shown at the left side of the rear-panel port row in the virtual diagram. It receives power from the device's approved adapter.
• WAN port: shown as a separate Ethernet port near the left side of the LAN group. It connects to the modem/ONT or ISP gateway.
• LAN ports: shown as a row labelled LAN 1, LAN 2, LAN 3, and LAN 4. They connect wired local devices or a switch.
• Status LEDs: shown along the top/front strip. These may represent power, internet/WAN, LAN link, and Wi-Fi activity.

Switch image identification

• Power port: shown on the left side of the rear-panel row. It supplies power to the switch.
• Ethernet ports: shown as numbered ports 1–8. Connect a different wired device to each port, or use a port for an uplink to the router.
• Link/activity LEDs: shown in the upper status area and commonly positioned close to each Ethernet port on real devices. They help show whether a link is active and whether data is moving.
• Console port: some managed switches have a separate console port for administration. The illustrated 8-port switch is a simple virtual example and does not show a console port; a basic unmanaged switch may not have one.

Image source suggestions

Cisco networking equipment and learning pages: https://www.cisco.com/
TP-Link official product pages and support: https://www.tp-link.com/
NETGEAR official product pages and support: https://www.netgear.com/

When using an online image, include the image page URL, manufacturer/model if known, and a short credit. Do not label a port as WAN or Console unless the device label or reliable product documentation confirms it. On many home routers, a console port is not present.

5. Scenario: Setting Up a Small Gaming Café

Imagine I am setting up a small gaming café for IPL streaming and multiplayer matches. The café has eight gaming PCs, one administrator PC, a network printer, and a smart TV used to show IPL matches. The café needs a stable internet connection, low local-network delay, and enough wired ports for all computers.

Which device connects first to the ISP line?

The ISP line first connects to the ISP-provided modem or optical network terminal (ONT), depending on the internet service. The Ethernet output of the modem/ONT then connects to the WAN/Internet port of the router. The router is the first of these two devices to connect to the ISP-facing connection because it manages the boundary between the ISP network and the café's private LAN. A switch should not normally replace the router at this point, because a basic Layer 2 switch does not provide the routing and gateway functions needed for ordinary internet sharing.

How would I connect eight gaming PCs?

I would connect one of the router's LAN ports to an Ethernet port on a gigabit (or faster, if needed) network switch. Then I would connect each of the eight gaming PCs to separate numbered ports on the switch using suitable Ethernet cables, preferably Cat 5e or Cat 6 for typical gigabit connections. The administrator PC, network printer, and any other wired devices could use remaining ports or a second switch if required. The router would provide IP addressing and internet routing, while the switch would carry local wired traffic between PCs and other LAN devices.

Why this design is suitable

• Stability: wired Ethernet is usually more consistent than Wi-Fi for fixed gaming PCs.
• Capacity: a switch provides enough ports to connect all eight PCs and other wired equipment.
• Local traffic: a switch can forward traffic between devices on the same LAN without sending every local exchange to the internet.
• Internet access: the router directs traffic to and from the ISP and can provide DHCP, NAT, and basic firewall functions.
• Easy troubleshooting: link/activity LEDs can help identify a disconnected cable or inactive port.

Example connection layout

ISP line → Modem / ONT → Router WAN port → Router LAN port → Gigabit switch → Gaming PCs 1–8
The router can also connect to the administrator PC or Wi-Fi access point. For eight PCs, choose a switch with at least eight usable ports plus an uplink/extra port as needed; a 16-port switch offers room for future devices.

Good setup practices

Use strong router administrator credentials, update router firmware, and enable a suitable firewall. Keep gaming PCs updated and use reputable security software. Use quality cables and label each cable at both ends (for example, PC-01 to Switch Port 1). Place networking equipment in a ventilated, secure location. If the café grows, consider managed-switch features such as VLANs and quality of service (QoS), configured by a qualified administrator.

Conclusion

A router and a switch are complementary devices. The router connects the local network to the ISP and forwards IP packets between networks. The switch connects wired devices on the LAN and forwards Ethernet frames using MAC-address information. Their ports and LEDs provide practical clues about power, connectivity, and activity. In a small gaming café, the ISP connection should reach the modem/ONT and then the router; the router's LAN connection should feed a switch, which connects the eight gaming PCs. This layout supports stable wired gaming, streaming, and straightforward expansion.

References / Further Reading

1. Cisco — Networking learning resources: https://www.cisco.com/
2. TP-Link — Router, switch, product, and support information: https://www.tp-link.com/
3. NETGEAR — Networking products and support information: https://www.netgear.com/
4. For exact port names and LED meanings, consult the user manual for the router or switch model being used.

End of Assignment